

PATRICK B. LUCAS

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Master Resume: Reference Repository (v25)

Full record of roles, tools, and achievements. Not for direct distribution, source material for tailored versions.

Operations leader with 10+ years of experience identifying broken processes and building permanent infrastructure to fix them. Specializes in ERP evaluation and implementation, custom tool development, demand forecasting, multi-site inventory management, and translating operational reality into system requirements. Builds tools that eliminate human memory and manual execution as dependencies, so processes produce consistent results regardless of who runs them.

CORE COMPETENCIES

ERP Systems & Implementation: ERPLY, JD Edwards EnterpriseOne (Time & Attendance / UAT), APlus/AS400 (order management, inventory, WMS), SAP S/4HANA (in progress), OnBase (PO/invoice linkage, AP support), XBE (quarry dispatch), HammerTech (safety inspections)	Process Analysis & Documentation: Business Requirements Gathering, UAT Planning & Defect Reporting
Full-Stack Tool Development: React, Node.js, PostgreSQL, Google Apps Script, Google Maps API, REST APIs	Supply Chain & Inventory Management: Demand Forecasting, S&OP, JIT Fulfillment, Multi-Site Materials Flow, Vendor Management
Procurement & PO Lifecycle Management: Landed Cost Analysis, RTV Processing, Cost Allocation & Margin Protection	Operations & Facilities Leadership: Union/Non-Union Team Management, OSHA Compliance, Payroll Administration, Receiving Operations

SYSTEMS & TOOLS BUILT

NEBP-ERP | React | Node.js | PostgreSQL | Role-based access control | PDF generation

Built to eliminate a single-person tribal-knowledge dependency: certain items (e.g., custom/non-standard brick with no formal item number) had no master record, requiring direct confirmation with one person for pricing every time, and purchasers not updating current costs led to frequent PO corrections. The prior plain-text email ordering process created real, costly errors, including full truckloads arriving at the wrong yard on drop-ship orders, requiring the load to be offloaded, reloaded, and reshipped to the correct location. Originated as a shared inventory tracker to replace this tribal knowledge, then expanded organically through a structured PO module, cost tracking, customer-specific pricing, and full AR/AP, covering the complete buy-to-cash and procure-to-pay cycle for sister company Northeast Building Products (NEBP), which had zero digital infrastructure. Mobile-responsive with role-based access. Built and functional; not yet deployed into daily production use. [Systems] [ERP Development] [Full-Stack]

Orgill Orderer | React | Node.js | PostgreSQL | CSV integration | Vendor portal upload

Rebuilt from a manual, click-through Orgill web ordering process into a full-stack procurement application. Diagnosed a structural landing-cost allocation problem where Orgill's flat \$55 fee, applied per PO rather than per order value, consistently inflated costs on the smallest of several simultaneous purchase orders while larger POs absorbed proportionally less under the 2%-of-value alternative; after an initial fix (receiving against vendor to disperse costs) proved too manual and error-prone for AP, consolidated all Orgill purchasing onto a single PO, eliminating the allocation problem entirely. Cross-references on-hand inventory against year-to-date and prior-year sales to recommend replenishment, while flagging any line with no purchase activity in 2+ years so it gets a deliberate manual review rather than an automatic reorder; the flag does not block the purchase; low-velocity items that anchor a higher-velocity consumable (e.g., paver roller frames, which sell rarely but support ongoing sales of the roller covers) still warrant ordering, and the tool puts the item in front of the user for that judgment call instead of letting it silently drop off a standard reorder report. Enforces full box quantity ordering to avoid split-box upcharges and applies independently adjustable shelf-life thresholds per product. Ingests Orgill's promotional pricing spreadsheet and cross-references it against sales velocity to flag which promo items are worth buying, outputting a CSV with promo codes embedded so discounts are correctly applied at order time. Cut ordering time from days (self-reported by the previous purchaser, who frequently redid orders due to manual errors) to a consistently timed 15 minutes across 880+ SKUs and \$70K+ in annual purchases, which changed the ordering cadence itself: the prior manual process supported only one or two large annual buys, while the automated process supports cutting a PO every 2-3 weeks during the selling season, with fewer, more thorough consolidated purchases intentionally retained in the off-season when there is more bandwidth to do them well. More frequent in-season ordering produces smaller receiving batches with fewer errors, tighter price verification per order, and faster response to actual demand shifts rather than reacting to trends months

late. When the previous purchaser left with no transition plan, the function was absorbed with no backfill and no disruption. [Procurement] [Automation] [Full-Stack] [Process Improvement]

O&G Material Calculator | *React* | *Vite* | *Node.js* | *PostgreSQL* | *Intranet-hosted* | *No cloud dependency*

Counter sales tool replacing pen-and-paper material estimation, which regularly produced three costly failure modes: undercounting forced customers into return trips for missing material, overcounting forced returns and restock fees on excess product, and paver square-footage miscalculation led to redone tickets and corrupted inventory records. Originated from a single complex bordered job where a customer's repeated spec changes forced the same manual math over and over, prompting a Google Sheets proof of concept (the O&G Cheat Sheet Master) before being rebuilt as a full-stack application. Models real material variables rather than flat area math: user-adjustable waste factors (typically 5% on straight edges, 10-12% on curved layouts), a piece-count formula that correctly distinguishes soldier coursing (long edge exposed, e.g., 3 pieces per linear foot on a 4x8 paver) from sailor coursing (short edge exposed, 1.5 pieces per linear foot on the same paver), a frequent source of counter miscalculation, and a polymeric sand quantity calculator based on average paver dimensions and joint width and depth. Covers SF pavers, piece pavers with soldier/sailor border math, cobblestone, Belgian block, wall block with linked cap pairs, CMU with mortar and core fill, and bulk materials, plus a Project Builder handling waste tiers, optional borders, and full underlayment specification, creating a natural upsell opportunity at the counter rather than leaving underlayment as an afterthought. All calculations round up, never down, structurally eliminating the undercounting failure mode. Dual output translates a sales-facing quantity (e.g., 391.4 sq ft) into an exact yard pull sheet (e.g., 3 pallets, 8 layers, 10.3 sq ft per layer), reducing pull errors and easing execution for yard staff. Password-protected admin panel for full catalog CRUD. Built in under one week. [Full-Stack] [Customer Service] [Process Improvement]

UPM Stock Tracker | *Google Sheets* | *Google Apps Script (automated email trigger)*

Inventory tracking system for Ultimate Paving Material (UPM), cold patch asphalt sold in winter when the asphalt plant is closed, a non-inventoried item with no visibility in the core ERP system. Separate inbound and outbound logs feed a stock page that calculates approximate on-hand tonnage in real time. An automated trigger sends an email alert when stock drops below a set threshold, closing a gap that previously had zero system-level visibility. [Inventory] [Automation] [Alerting]

Fork Truck Hours Tracker | *Google Sheets*

Daily equipment-hour logging tool for forklifts and front-end loaders. Yard personnel report each machine's current hour meter reading; the tool rolls the prior day's reading forward and calculates the daily delta, rounded to the nearest quarter hour. Hours are then assigned to operators in JDE's time clock system, supporting maintenance scheduling based on actual usage and providing data-backed justification for which equipment stays on site versus gets reallocated or retired. [Equipment Tracking] [JDE] [Maintenance]

Additional Operational Tools Library | *Google Sheets* | *Google Apps Script*

Maintains a working library of 10+ additional operational tools built to close gaps as they surface in daily operations: a digital truck-scale weight calculator replacing an illegible shared paper log; a bin map and tonnage-conversion reference for bulk aggregate products; a vendor-specific reorder and truck-balancing calculator for Chicopee block; a weekly-updated per-vendor landing cost calculator; a staff scheduling tool; and a bulk PO management sheet. [Automation] [Inventory] [Logistics]

Lehigh Drop Ship Pricing Calculator | *Google Sheets* | *Excel formulas (VLOOKUP/XLOOKUP)*

Assigned directly by the EVP, over the existing purchasing team, to build a drop-ship pricing tool after sales staff struggled to price resale accounts accurately. Built a dynamic landed-cost calculator that computes per-bag freight cost using customer-specific haul rates divided by DOT-compliant truck capacity, which varies by destination state, then applies a configurable markup to generate accurate, compliant sale pricing per customer and product. [Pricing] [DOT Compliance] [Leadership Visibility] [Automation]

Vendor Order Recommendation Engine | *Google Sheets* | *Google Apps Script*

Built after growing tired of manually re-running the same demand math on every load across six hardscape vendors (CST, Unilock, Techo-Bloc, Nicolock, Belgard, Thinstone), each run through a dedicated custom menu with vendor-specific processing. For each vendor, automatically cleans raw data by removing discontinued items and non-stock products with no sales activity, injects custom formulas calculating per-pallet quantities, on-hand-versus-target fill percentage, and pallets needed to reach historical demand levels, then sorts output by greatest need. Output feeds directly into the truck-load planning system below to convert recommendations into DOT-compliant, ready-to-cut purchase orders. [Procurement] [Automation] [Vendor Management]

Truck Load & DOT Compliance Planner | *Google Sheets* | *Google Apps Script*

Built after growing tired of manually re-running the same weight math on every load, submitting a PO only to have it rejected as overweight, underweight, or needing revision. The CST and Unilock tabs use a full grid of live per-pallet weights by material and color, cross-referencing them in real time against truck payload capacity to flag what can still be

added without exceeding DOT weight limits. Separate tabs for Lehigh cement, Quikrete, and block use simpler load-balancing calculators with conditional formatting. Eliminates the submit-reject-revise cycle entirely, preventing overweight loads, short-loaded trucks, and costly PO revisions. [Logistics] [Compliance] [Automation]

Open PO by Vendor Report Tool | Google Sheets | Google Apps Script | Reporting system integration

Pulls the master open PO report from the reporting system, strips noise including pallet charges and landing costs, and breaks the master report into individual yard-level views by vendor. Enables cross-referencing of open POs against vendor records to catch discrepancies, specifically cases where O&G shows an order open but the vendor has already closed it. [Procurement] [Reporting] [Automation]

Open Orders Aging Report Tool | Google Sheets | Google Apps Script

Runs open orders greater than 30 days and splits output by sales rep, each on their own page. Used to push sales reps to either convert allocated material into completed sales or release it back to available inventory. Recovers locked stock and improves inventory visibility across the sales pipeline. [Sales Ops] [Reporting] [Automation]

ERPLY UX & Process Evaluation | Structured UAT documentation

Authored a detailed structural evaluation of ERPLY's point-of-sale workflow during the company's enterprise ERP rollout. Identified issues invisible to a systems-only review, including missing unit-of-measure display at the item level, barcode scanning not routing to the default search field, uncontrolled duplicate customer record creation, and unrestricted access to sensitive account fields. Recommendations were grounded in day-to-day counter operations experience rather than abstract system requirements. One of two contributors alongside a purchaser focused on inventory-side feedback. [Systems Analysis] [UAT] [ERPLY]

ThriveMetrics | React PWA | Supabase (Auth/DB/Edge Functions) | Vercel | 6 external integrations

Personal fitness PWA consolidating workout, nutrition, weight, and sleep data into a single dashboard with AI coaching, deployed live at peloton-dashboard-kappa.vercel.app. Originally built in Google Sheets, fully migrated to a modern stack (React, Supabase, Vercel). Integrates Strava, Fitbit, USDA FoodData Central and OpenFoodFacts, and Gmail for automated report delivery. Generates a weekly AI coach report using dual parallel Claude calls, and a monthly insurance compliance report automating a real regulatory reporting requirement. Demonstrates full-stack development, third-party API integration, and applied AI/LLM integration outside a work context. [Full-Stack] [AI/LLM Integration] [Personal Project]

PROFESSIONAL EXPERIENCE

Senior Operations & Systems Manager | O&G Industries | Waterbury, CT **2019 – Present**

Connecticut's largest privately-held construction company, a national Top 400 Contractor, founded 1923. 7-location building materials division; managed facility supports \$10M+ annual revenue.

Own daily facility operations: coordinate dispatch and delivery, manage vendor relationships and PO cutting, handle receiving (99.9% of facility volume), resolve customer and product issues in real time, and serve as the go-to operational knowledge base across the wider division. Regularly sought out across the division as the resource for JDE time-clock/UAT questions, ERPLY functionality, Excel troubleshooting, and general technical support, work that falls outside formal role scope but reflects the operational and systems reputation built over 10+ years. Detailed achievements and systems work below are built on top of this daily operational base, not in place of it.

ERP Implementation & Business Analysis

- Assigned by the Executive Vice President to lead operational evaluation of a division-wide ERPLY rollout, based on recognized cross-functional operations and systems expertise; delivered structured evaluation across POS, back office, ordering, inventory, receiving, and vendor management directly to senior leadership [ERPLY] [Systems Analysis] [Leadership Visibility]
- Authored 25-page ERP implementation roadmap documenting 40+ system gaps, UX deficiencies, and process improvement recommendations across POS, procurement, inventory, ordering, and security functions [ERPLY] [Systems Analysis] [Documentation]
- Delivered 16-page follow-up markup document after second trial evaluating administrative improvements and identifying remaining gaps, demonstrating structured iterative review process [ERPLY] [Systems Analysis]
- Designated ERPLY super user by the Inventory Control Manager and IT Development Manager at the EVP's direction, with full admin functions and manager-tier system access [ERPLY] [Systems Admin]
- Began testing JDE EnterpriseOne time and attendance features informally after demonstrating strong command of the system during basic training; this expanded organically into ongoing UAT responsibility as new features

required testing. Part of the initial two-location pilot (with Torrington) before division-wide rollout to all locations; remain the only location conducting ongoing UAT and defect documentation [JDE] [UAT]

- Identified and documented multiple JDE defects across two release cycles including non-quarter-hour increment validation failure, ETO stacking above full shift hours, cost center regression that reappeared after each new build, missing Pay Type population in Manager Approval screen, and absent ESS confirmation feedback [JDE] [UAT] [Defect Documentation]
- Proposed workflow consolidation to reduce ETO entry from two screens to one, limiting pay type options by employee classification to prevent errors and reduce training burden [JDE] [Process Improvement]
- Collaborated with JDE Application Manager, Business Analyst, Developer, IT, and Payroll team on system configuration and testing [JDE] [Cross-Functional]
- Serve as an informal cross-location systems resource on APlus (order management, inventory, staged pick/virtual dock behavior): resolved a stuck partial-shipment pick issue for the Assistant Manager at the Stamford location via remote screen share on a weekend with no support staff available, correcting both the immediate stuck ticket and the underlying picking procedure for future partial shipments; separately, Credit department staff have repeatedly routed customer account configuration questions directly to me, including a house-account emailed-invoice setup and an account settings change a 25-year salesperson and a 15-year assistant manager both believed was not possible [APlus] [Cross-Functional] [Systems Expertise]
- Maintain PO-to-invoice linkage in OnBase, updating records with JDE PO numbers and uploading invoices and packing slips to support AP reconciliation and Credit's billing process [OnBase] [AP/Credit]
- Order bulk quarry material (stone dust, trap rock, processed stone, quarry blend, 1/2-inch native, sand) across four quarry sources by product and delivery timing, and schedule triaxle deliveries from yard to job site, using XBE dispatch software [XBE] [Procurement] [Logistics]
- Discovered DYMO label printers could generate scannable barcodes and began labeling yard hook hangers with the product assigned to each location, turning restocking into a scan-and-match action instead of relying on staff memory [Process Improvement] [Inventory]

Procurement & Vendor Management

- Coordinate across facilities on multi-site deliveries, splitting orders and working directly with other location managers and dispatch to determine optimal sourcing [Logistics] [Cross-Functional]
- Rebuilt entire procurement operation for 880+ SKUs after critical purchaser departure with no transition plan; function absorbed with no backfill and no disruption to operations [Procurement] [Ownership]
- Maintain active ordering and availability accuracy across roughly 1,100+ SKUs spanning five vendor relationships (CST/Pavestone: 111, Unilock: 180, Quikrete: 13, Chicopee: 15, Orgill: 813+), with sole ownership of the largest single catalog [Procurement] [Vendor Management]
- Monitor vendor cost changes and flag margin threshold breaches to purchasing, ensuring sale pricing gets remapped before margin erosion goes unnoticed [Procurement] [Financial Analysis]
- Identified and resolved \$55-per-unit landing cost allocation error that had been invisible in the receiving process [Procurement] [Process Improvement] [Financial Analysis]
- Reduced PO cycle time from 3-5 days to under 15 minutes through full-stack procurement automation integrating live inventory, sales history, shelf-life parameters, and case pack compliance [Automation] [Procurement]
- Built vendor order optimization tool integrating load checks and weight validation to prevent costly delivery rejections [Automation] [Logistics]
- Orchestrated \$850K+ multi-vendor construction logistics across 35+ truckloads and 2 major projects with zero delays attributable to load planning or customer communication; built custom tracking systems to overcome ERP limitations with generic item codes [Logistics] [Vendor Management]
- Own the full procure-to-pay cycle for large-scale multi-release orders: cut POs against contract terms, coordinate staged releases with dispatch and vendor, reconcile vendor invoices against signed proof-of-delivery, assign landed cost, and track running balance by product line in a custom-built spreadsheet system [Procurement] [Project Management]
- Manage 25+ vendor accounts; process return POs, calculate landed cost per unit, collect credits, and initiate reimbursement requisitions [Vendor Management] [Procurement]

- Set contracted and job-specific customer pricing directly, a function absorbed after repeated pricing gaps from sales created recurring credit and rebill cleanup; handling it directly eliminates that rework [Pricing] [Process Improvement]
- Continuously apply root-cause analysis to recurring operational failures rather than treating them as isolated incidents [Process Improvement]
- Brought in directly by ownership (Aaron and Jeremy) to diagnose recurring pallet inventory losses; identified five distinct, compounding failure points rather than a single cause: chargeable pallets physically dispersed across the yard rather than centralized like a standard SKU, ambiguity between chargeable and non-chargeable pallets, pallets not consistently treated as inventory at the point of breakage or use, an RTV/credit process that leaves pallets counted as on-hand while physically already returned to the vendor until a credit memo is finalized, and occasional receiving-side errors generally caught by AP; also evaluated and pushed back on a proposed blanket fix (charging for all pallet types), identifying it would roughly quadruple the units requiring physical count (hundreds to 1,000+) without resolving the underlying dispersion and tracking problems, and would create new customer disputes over damage on non-durable pallet types vendors will not accept returns on; delivered a training- and process-based recommendation instead, covering pallet identification criteria, cross-department credit notification, and driver-signed return documentation [Process Improvement] [SOP Design] [Root Cause Analysis]
- Example: diagnosed \$2,220 in recoverable pallet write-offs at East Lyme location as process failure rather than shrinkage; designed 7-step pallet return SOP covering vendor notification, yard staging, dual-copy BOL execution, AP notification, credit tracking, and PO closure [Process Improvement] [SOP Design]
- Handle nearly all receiving for the facility (99.9% of volume), closing the loop on the procure-to-pay cycle from PO through goods receipt [Procurement] [Operations]

Inventory & Supply Chain Management

- Manage \$2M inventory portfolio across Waterbury facility supporting \$10M+ annual revenue [Inventory]
- Built automated delivery routing matrix using Google Maps API covering 7 facilities and 150+ CT towns [Automation] [Logistics]
- Oversee multi-location physical inventory efforts across 4 yards [Inventory]
- Conducted forensic inventory investigations combining live recounts, ERP history audits, receiving logs, and sales data to identify discrepancies and recover financial exposure [Inventory] [Data & Analytics]
- Cross-reference system timestamps (JDE, APlus) against video surveillance footage to build a complete evidentiary picture for inventory discrepancies and workplace accident reports, backing paper trails with visual confirmation [Data & Analytics] [Safety]
- Reviews sales trends and surplus inventory with sales staff to align purchasing and stock levels with current demand [Inventory] [Cross-Functional]

Operations & Facilities Management

- Administer weekly payroll for 8+ staff: correct employee clock-in/out errors, manually input driver hours from physical logs, log machine hours for assigned trucks, Moffett, forklifts, and front-end loader, and review accumulated hours to flag equipment unsafe for continued use; approve hours weekly through ESS and manage ETO, jury duty, and bereavement time categorization; also manage utility accounts, vendor services, and coordination of internal/external maintenance [Operations] [Leadership] [Payroll]
- Coordinated with AP/AR/Credit departments to resolve billing discrepancies, process vendor payments, and maintain relationships with 25+ supplier accounts [Vendor Management] [Finance]
- Acted as on-site first responder for IT support including connectivity issues, network outages, and hardware setup [IT Support]
- Maintained OSHA compliance through monthly inspections, red-tagging unsafe equipment, incident documentation, and HammerTech reporting; multiple years with zero lost-time incidents [Safety] [Compliance]
- Proactively identify and resolve physical hazards before they cause incidents: had exposed conduit surfacing through yard asphalt repaired before it could be caught by a snowplow, identified a retaining wall at risk of structural failure, and traced asphalt fragments appearing in material bins back to pavement deterioration requiring repair [Safety] [Initiative]
- Championed a proactive safety culture beyond formal compliance [Safety] [Leadership]
- Led hiring, cross-training, and performance management for union and non-union personnel; advanced sales associate to Assistant Operations Manager through structured mentorship [Leadership] [Talent Development]

Promoted from Inventory Control Specialist.

- Managed daily warehouse operations at high-volume regional consolidation facility distributing tools and specialty items to 7 company locations [Operations]
- Took over stucco tinting operations for NY Stucco following Parex's acquisition, after diagnosing a root-cause formula mix-up where staff could apply the wrong base preset across product lines, leading to failed applications and, in some cases, O&G covering customer labor costs; restructured formulas by product and base and built a full mixing protocol; significantly reduced color mismatches [Process Improvement] [Cross-Functional] [Vendor Management]
- Identified that warehouse layout was organized on retail-exposure logic rather than pick-efficiency logic; quantified the cost of the mismatch on a single high-velocity SKU (13.4 miles of unnecessary annual forklift travel) as a proof point, then applied 80/20 Pareto analysis to systematically reposition SKUs by velocity [Process Improvement] [Data & Analytics]
- Introduced structured labeling and barcoding system using APlus WMS and DYMO printers; improved material traceability and reduced yard search time [Process Improvement] [APlus]
- When Kara and Bob asked managers in a group meeting for suggestions to improve the company, and the room otherwise had none, authored a 7-page proposal spanning warehouse slotting and heat-map analysis, product labeling and identification systems, seasonal demand forecasting, transfer consolidation, process automation opportunities, internal and external competitive benchmarking, and organizational culture frameworks; presented it directly to Kara and Bob in a follow-up meeting, which resulted in promotion [Strategic Thinking] [Leadership]
- Trained and cross-developed 10+ employees across yard, counter, and logistics roles [Talent Development]
- Prepared shipment documentation for 3 drivers based at the Bridgeport facility, plus additional drivers from other yards routing through Bridgeport; coordinated up to 20 deliveries daily during peak periods [Logistics]

Inventory Control Specialist | O&G Industries | Stamford, CT**2015 – 2017**

- Managed inbound/outbound gate operations; verified Bills of Lading, directed unloading, performed final load verification for outbound deliveries [Logistics]
- Oversaw dedicated warehouse containing PVC fittings, drainage components, block materials, and water mitigation products [Inventory]
- Conducted full physical inventory counts and provided availability reports to support system reconciliation [Inventory]
- Handed a broken ratchet and a hand wrench and told to assemble wheelbarrows, with no workstation of any kind provided; built a complete mobile assembly station from scratch using a discarded crate and a pallet jack, then sourced an impact driver to replace the manual tools; increased assembly rate roughly 10x, from a 2-per-day baseline to 20+ per day [Process Improvement] [Initiative]

EDUCATION

B.S. Business Management | Southern New Hampshire University, 2018 | GPA: 3.98

Major: Operations & Project Management | Minor: Information Technology

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- SAP S/4HANA, Supply Chain & Operations Management (In Progress, learning.sap.com)
- Google Data Analytics Professional Certificate: Excel, SQL, R, Tableau (Coursera, 2023)
- ISO 9001:2015 Quality Management Systems (Alison, 2025), note: low-value credential, exclude from targeted resumes
- AS9100D Aerospace Quality Management Systems (Alison, 2025), note: low-value credential, exclude from targeted resumes
- OSHA Forklift Certification (Current), include only for warehouse/facility-management-targeted roles